LAPORAN PRAKTIKUM
TEKNOLOGI IoT II
4 dan 5



DISUSUN OLEH:

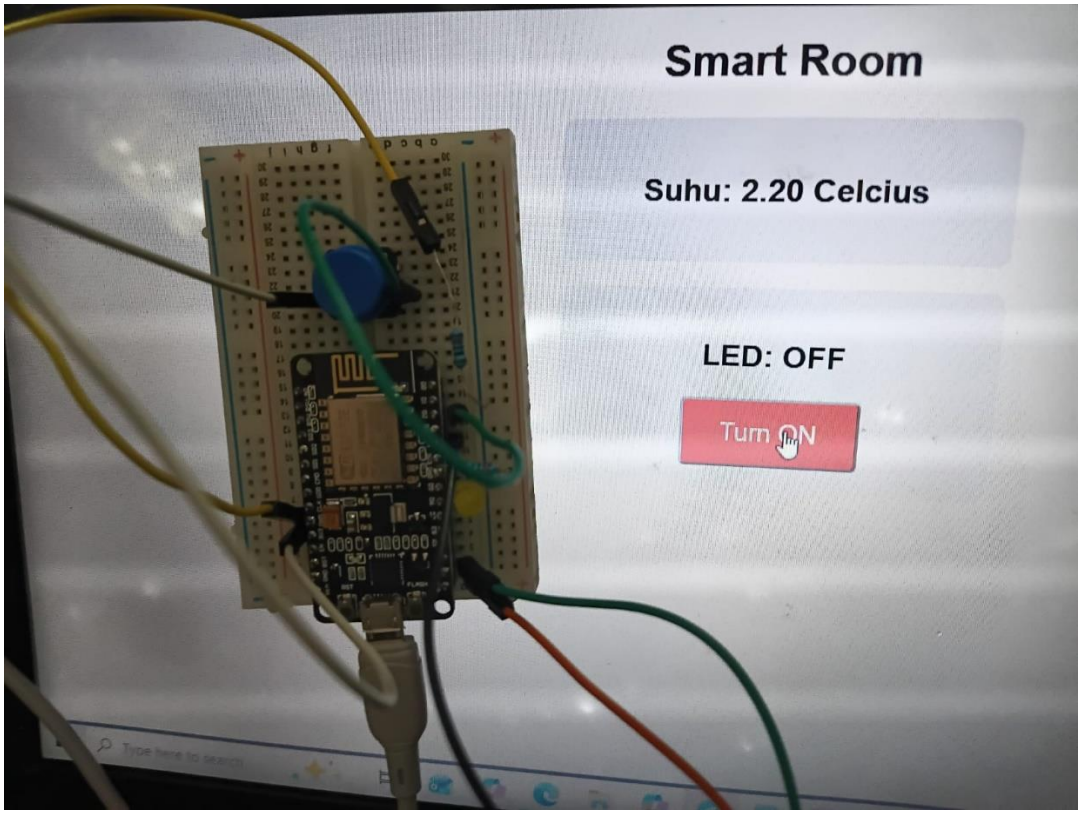
Nama : Zahara Ummu Abdillah
NPM : 25782058
Kelas :3B

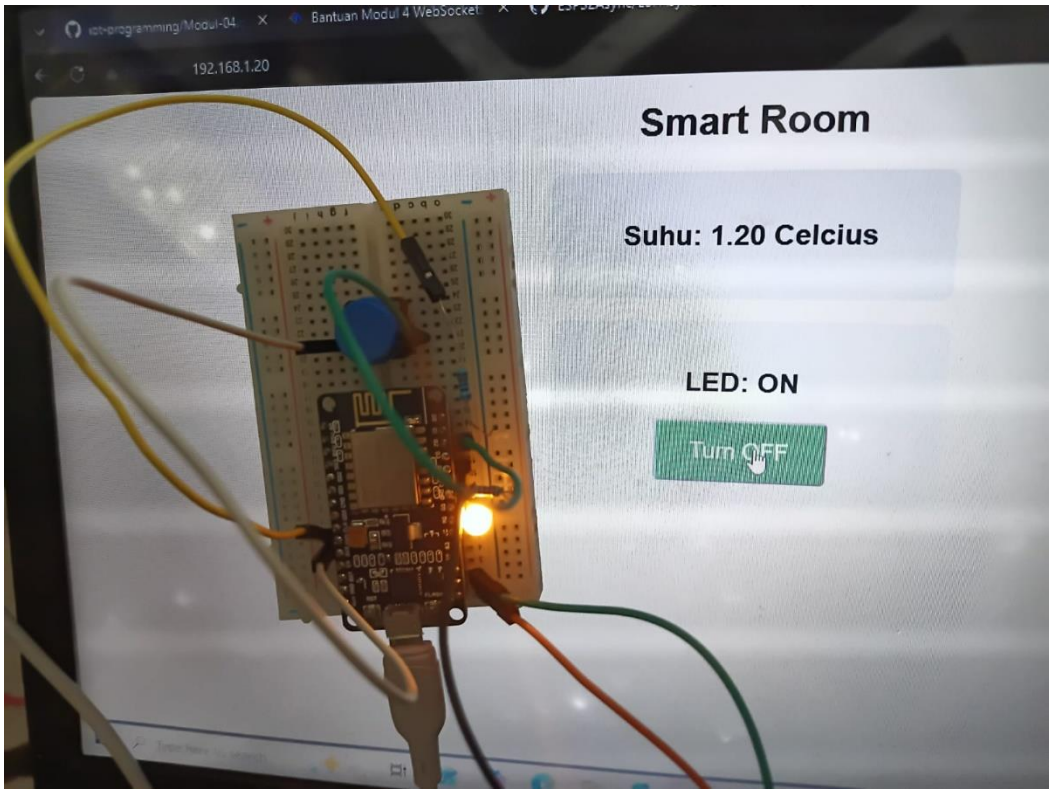
Dosen Pengampu:
Dr. Septafiansyah Dwi Putra, S.T., M.T.
Agus Ambarwari, S.Pd., M.Kom.
Andi Saptono, S.Kom

TEKNOLOGI REKAYASA INTERNET
JURUSAN TEKNOLOGI INFORMASI
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2026/2027

➢ HASIL PELAKSANAAN PRAKTIKUM

4.5.1 Praktikum 1: Real-Time Dashboard & Sinkronisasi Tombol Fisik





4.6 Latihan

Latihan 1 (Teori Jaringan): Jelaskan dengan kata-kata Anda sendiri, mengapa teknologi koneksi WebSockets jauh lebih hemat konsumsi data/bandwidth internet dan tidak membuat layar berkedip jika dibandingkan dengan metode Meta Refresh HTTP HTML yang Anda gunakan pada Tugas Praktikum Modul 3?

Jawab: Berdasarkan pengamatan saya, WebSockets jauh lebih efisien dalam penggunaan data dibandingkan HTTP Meta Refresh. Pada metode Meta Refresh, peramban harus memuat ulang seluruh halaman beserta komponennya secara berkala, yang memicu pemborosan data serta efek flickering (layar berkedip). Sebaliknya, dengan WebSockets, koneksi antara peramban dan NodeMCU tetap terbuka untuk komunikasi dua arah. Server cukup mengirimkan data yang berubah (seperti suhu atau status LED) dalam format JSON mikro. JavaScript kemudian memperbarui tampilan secara instan melalui manipulasi DOM tanpa refresh halaman.

Latihan 2 (Analisis Waktu/Timer): Apa bahaya sistematis yang akan terjadi pada komunikasi WebSockets (reaksi antara browser dan NodeMCU) jika Anda bersikeras nekat memanggil sintaks delay(3000) di dalam siklus fungsi void loop() pada draf arsitektur kompleks ini?

Jawab: Jika kita menggunakan delay(3000) di dalam loop(), CPU NodeMCU akan dibekukan (sleep) sepenuhnya selama 3 detik. Akibatnya:

1. NodeMCU tidak bisa merespons input tombol fisik saat tombol ditekan di dalam jeda3 detik tersebut.
2. Koneksi WebSocket akan terganggu/lag, atau bahkan terputus (timeout) karena server tidak sempat melayani lalu lintas jaringan (cleanup/event loop).
3. Pengalaman pengguna (user experience) menjadi sangat lambat dan tidak real-time.

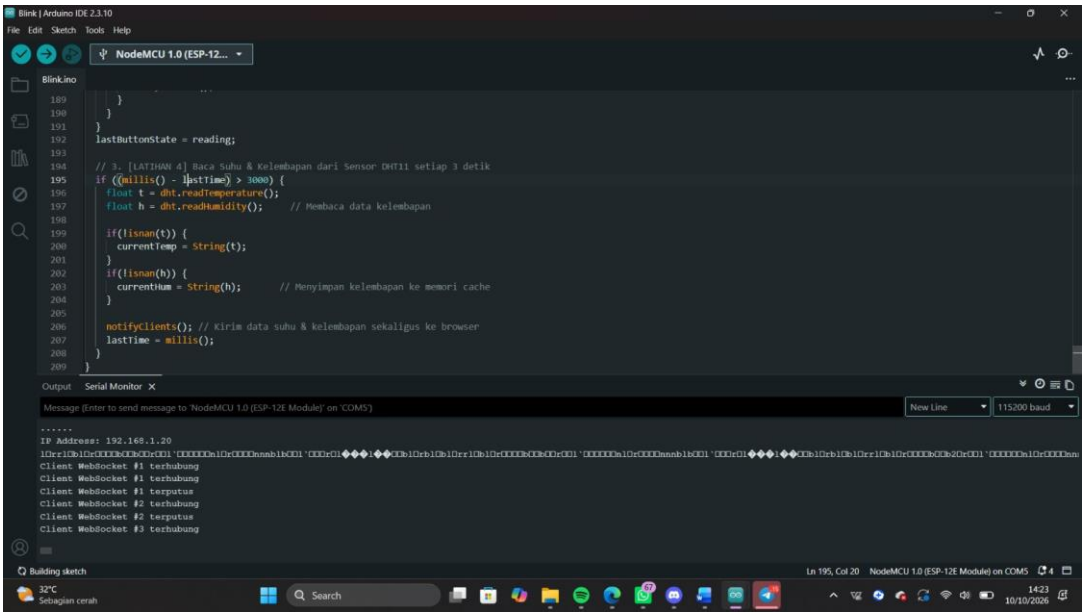
Latihan 3 (Modifikasi Front-End): Analisislah fungsi JavaScript pada blok // Merombak UI Tombol & Teks sesuai status hardware. Nama ID elemen HTML () apa saja yang menjadi target injeksi/pengubahan teks oleh script yang ditulis oleh Anda?

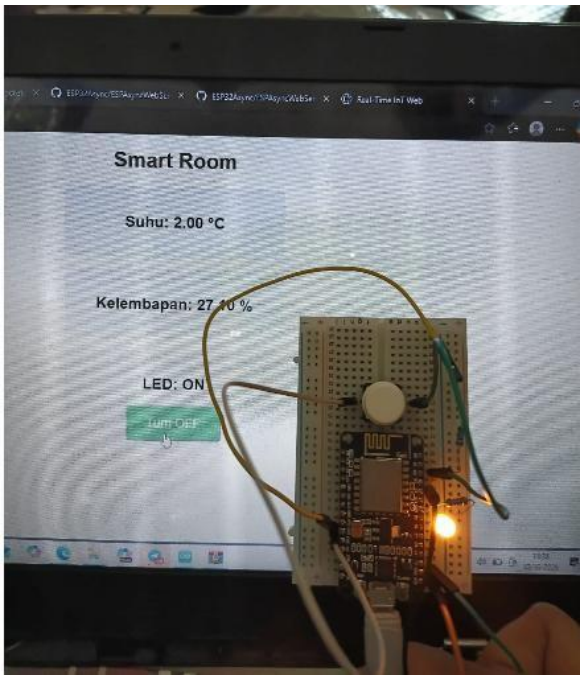
Jawab: ID elemen HTML yang menjadi target pengubahan/injeksi oleh JavaScript pada fungsi onMessage() adalah:

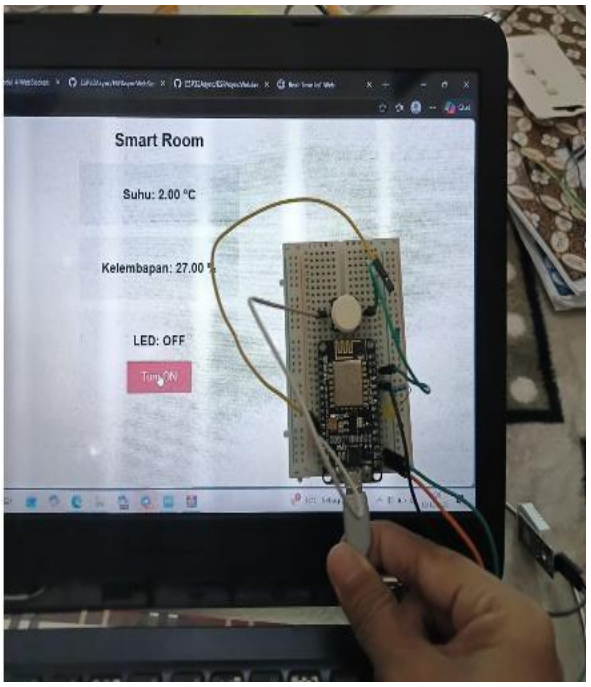
1. tempValue: Menerima injeksi nilai suhu (dataObj.suhu).
2. ledStatus: Menerima injeksi teks status LED (ON atau OFF).
3. toggleBtn: Menerima perubahan teks tombol (Turn OFF / Turn ON) dan perubahan warna kelas CSS (btn-on / btn-off).

Latihan 4 (Pengembangan Telemetri Sensor - Kelembapan): Pada Praktikum 1, antarmuka web hanya menyajikan data suhu (temperature). Modifikasilah program agar sistem dapat membaca dan menampilkan data kelembapan (humidity) secara real-time via WebSockets:

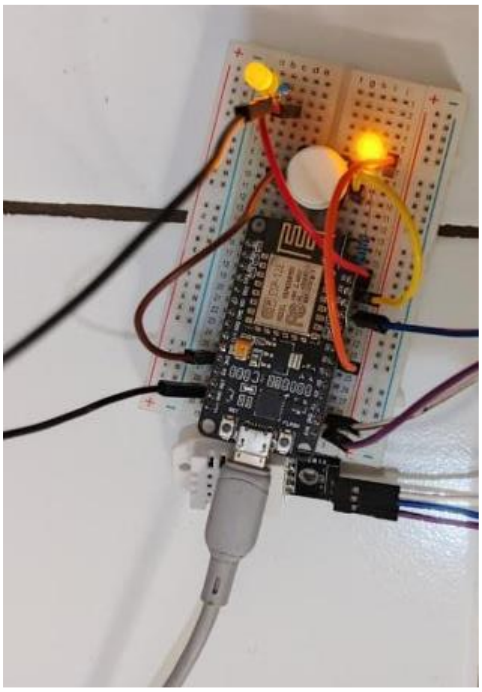
Jawab:

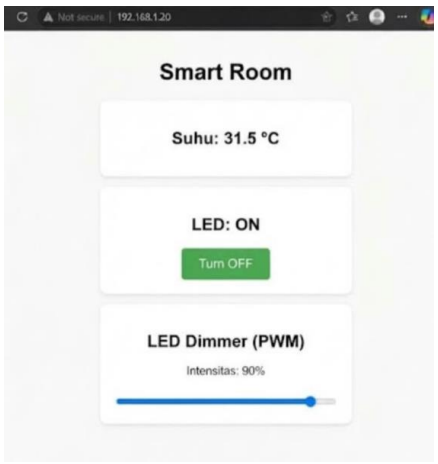






4.7 Tugas: Sistem Kendali Intensitas Cahaya (PWM Slider) via WebSockets

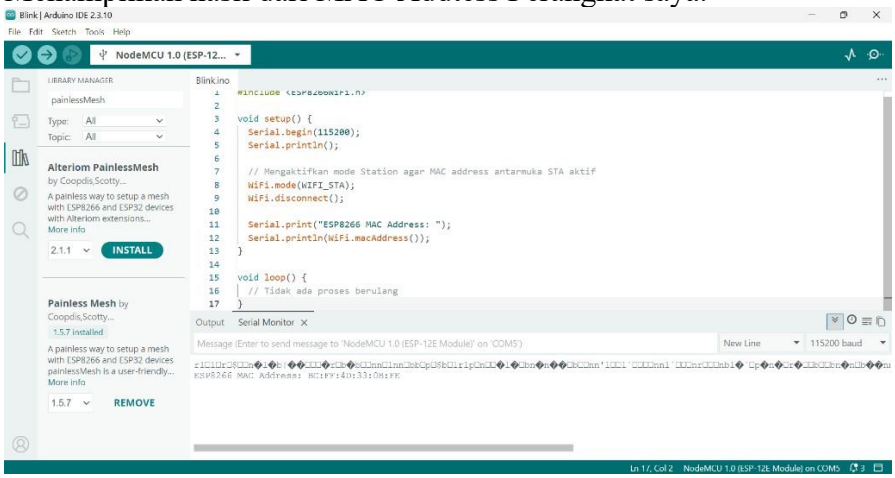






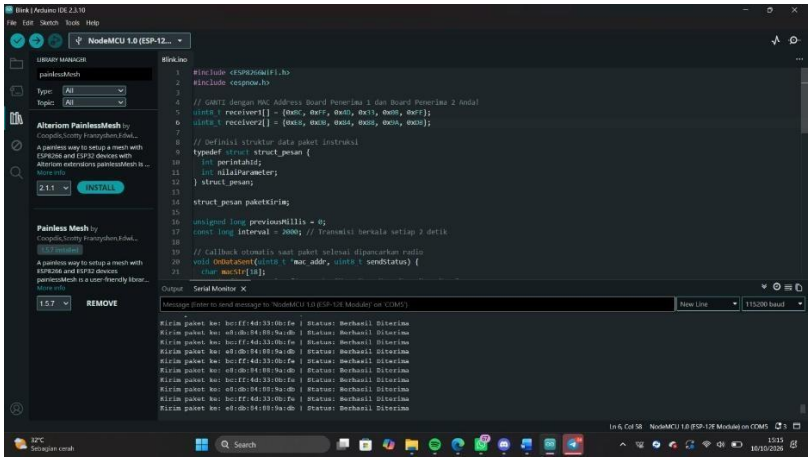
5.5.1 Praktikum 1: Identifikasi MAC Address Perangkat NodeMCU

Menampilkan hasil dari MAC Addtess Perangkat saya:

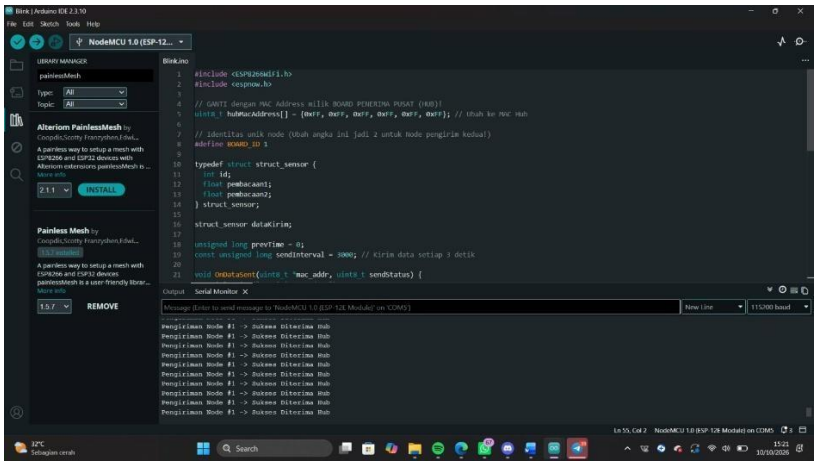


5.5.2 Praktikum 2: Komunikasi Nirkabel One-to-Many (ESP-NOW) Sebagai Penerima:

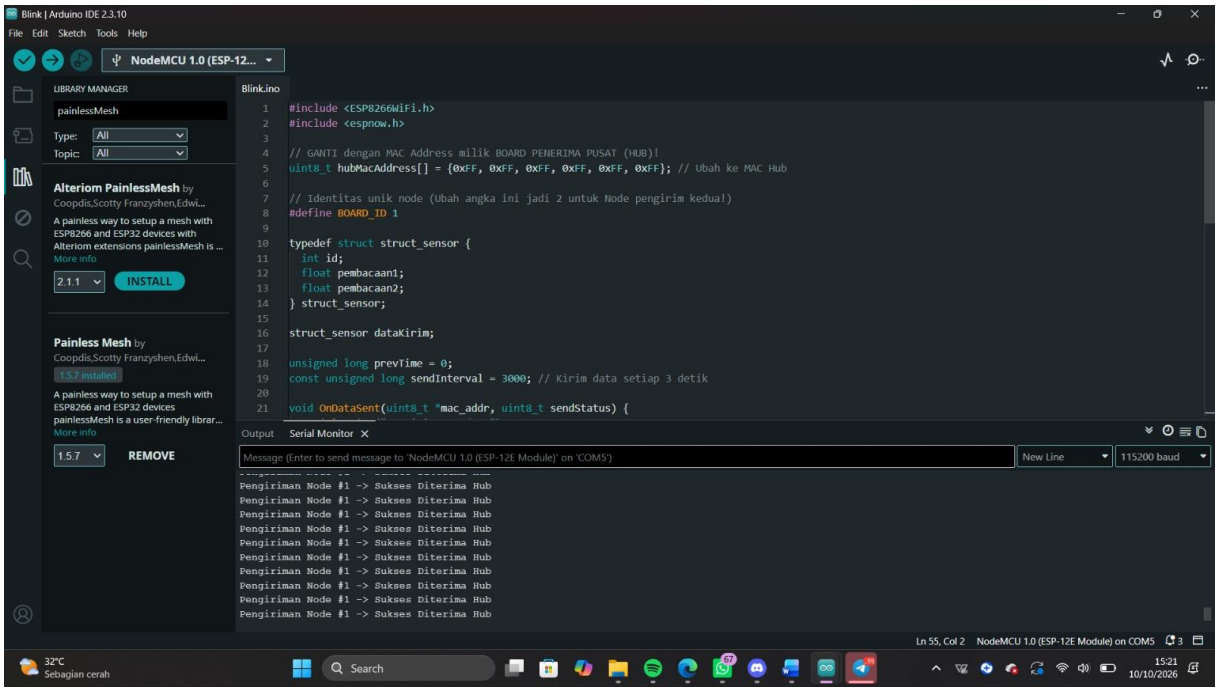
PENGIRIM :



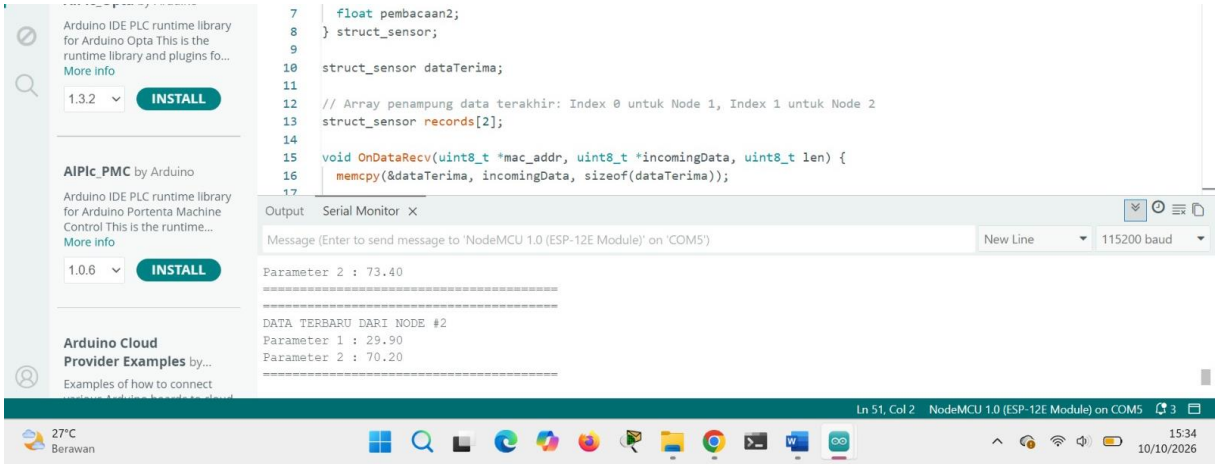
PENERIMA:



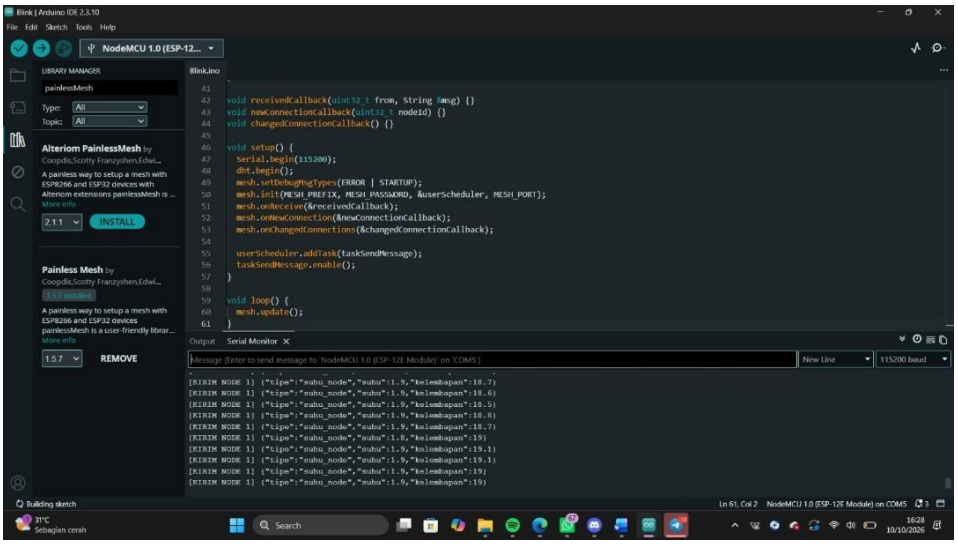
5.5.3 Praktikum 3: Pengumpulan Data Telemetri Many-to-One (ESP-NOW)



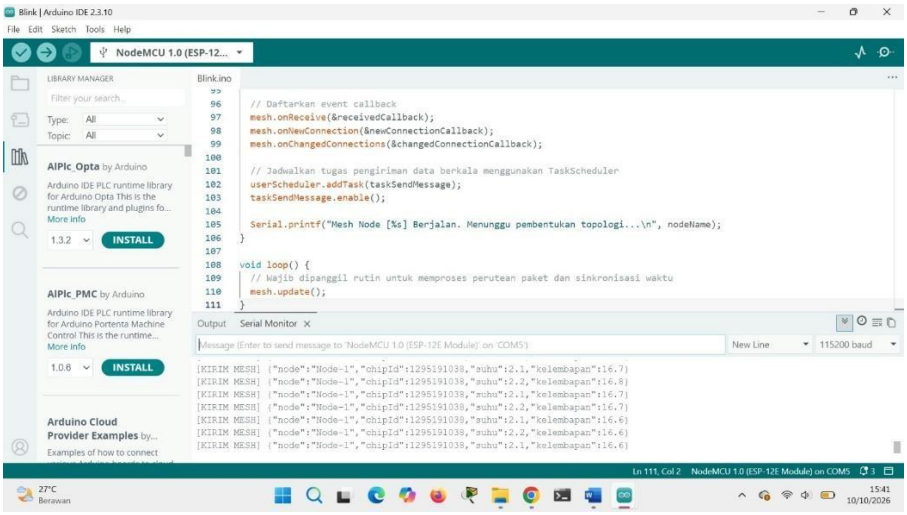
5.5.4 Praktikum 4: Jaringan Sensor Ad-Hoc Multi-Hop (painlessMesh)



NODE :


Blink | Arduino IDE 2.3.10
File Edit Sketch Tools Help
NodeMCU 1.0 (ESP-12...
LIBRARY MANAGER
painlessMesh
Type: All
Topic: All
Alteriom PainlessMesh by Coopdis,Scotty Franzyshen,Edwi...
A painless way to setup a mesh with ESP8266 and ESP32 devices with Alteriom extensions painlessMesh is ...
More info
2.1.1 INSTALL
Painless Mesh by Coopdis,Scotty Franzyshen,Edwi...
1.5.7 installed
A painless way to setup a mesh with ESP8266 and ESP32 devices painlessMesh is a user-friendly librar...
More info
1.5.7 REMOVE
Blink.ino
41
42 void receivedCallback(uint32_t from, String &msg) {}
43 void newConnectionCallback(uint32_t nodeId) {}
44 void changedConnectionCallback() {}
45
46 void setup() {
47 Serial.begin(115200);
48 dht.begin();
49 mesh.setDebugMsgTypes(ERROR | STARTUP);
50 mesh.init(MESH_PREFIX, MESH_PASSWORD, &userScheduler, MESH_PORT);
51 mesh.onReceive(&receivedCallback);
52 mesh.onNewConnection(&newConnectionCallback);
53 mesh.onChangedConnections(&changedConnectionCallback);
54
55 userScheduler.addTask(taskSendMessage);
56 taskSendMessage.enable();
57 }
58
59 void loop() {
60 mesh.update();
61 }
Output Serial Monitor
Message (Enter to send message to 'NodeMCU 1.0 (ESP-12E Module)' on 'COM5')
New Line
115200 baud
[KIRIM NODE 1] {"tipe":"suhu_node","suhu":1.9,"kelembapan":18.7}
[KIRIM NODE 1] {"tipe":"suhu_node","suhu":1.9,"kelembapan":18.6}
[KIRIM NODE 1] {"tipe":"suhu_node","suhu":1.9,"kelembapan":18.5}
[KIRIM NODE 1] {"tipe":"suhu_node","suhu":1.9,"kelembapan":18.8}
[KIRIM NODE 1] {"tipe":"suhu_node","suhu":1.9,"kelembapan":18.7}
[KIRIM NODE 1] {"tipe":"suhu_node","suhu":1.8,"kelembapan":19}
[KIRIM NODE 1] {"tipe":"suhu_node","suhu":1.9,"kelembapan":19.1}
[KIRIM NODE 1] {"tipe":"suhu_node","suhu":1.9,"kelembapan":19.1}
[KIRIM NODE 1] {"tipe":"suhu_node","suhu":1.9,"kelembapan":19}
[KIRIM NODE 1] {"tipe":"suhu_node","suhu":1.9,"kelembapan":19}
Building sketch
Ln 61, Col 2 NodeMCU 1.0 (ESP-12E Module) on COM5
31°C Sebagian cerah
Search
16:28 10/10/2026

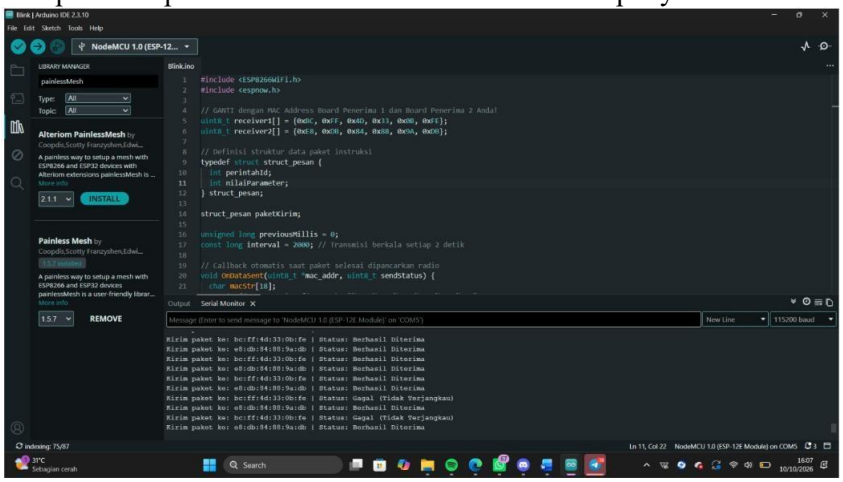
MESH :


Blink | Arduino IDE 2.3.10
File Edit Sketch Tools Help
NodeMCU 1.0 (ESP-12...
LIBRARY MANAGER
Filter your search...
Type: All
Topic: All
AIPlc_Opta by Arduino
Arduino IDE PLC runtime library for Arduino Opta This is the runtime library and plugins fo...
More info
1.3.2 INSTALL
AIPlc_PMC by Arduino
Arduino IDE PLC runtime library for Arduino Portenta Machine Control This is the runtime...
More info
1.0.6 INSTALL
Arduino Cloud Provider Examples by...
Examples of how to connect
Blink.ino
96 // Daftarkan event callback
97 mesh.onReceive(&receivedCallback);
98 mesh.onNewConnection(&newConnectionCallback);
99 mesh.onChangedConnections(&changedConnectionCallback);
100
101 // Jadwalkan tugas pengiriman data berkala menggunakan TaskScheduler
102 userScheduler.addTask(taskSendMessage);
103 taskSendMessage.enable();
104
105 Serial.printf("Mesh Node [%s] Berjalan. Menunggu pembentukan topologi...\n", nodeName);
106 }
107
108 void loop() {
109 // Wajib dipanggil rutin untuk memproses perutean paket dan sinkronisasi waktu
110 mesh.update();
111 }
Output Serial Monitor
Message (Enter to send message to 'NodeMCU 1.0 (ESP-12E Module)' on 'COM5')
New Line
115200 baud
[KIRIM MESH] {"node":"Node-1","chipId":1295191038,"suhu":2.1,"kelembapan":16.7}
[KIRIM MESH] {"node":"Node-1","chipId":1295191038,"suhu":2.2,"kelembapan":16.8}
[KIRIM MESH] {"node":"Node-1","chipId":1295191038,"suhu":2.1,"kelembapan":16.7}
[KIRIM MESH] {"node":"Node-1","chipId":1295191038,"suhu":2.2,"kelembapan":16.7}
[KIRIM MESH] {"node":"Node-1","chipId":1295191038,"suhu":2.1,"kelembapan":16.6}
[KIRIM MESH] {"node":"Node-1","chipId":1295191038,"suhu":2.2,"kelembapan":16.6}
[KIRIM MESH] {"node":"Node-1","chipId":1295191038,"suhu":2.1,"kelembapan":16.6}
Ln 111, Col 2 NodeMCU 1.0 (ESP-12E Module) on COM5
27°C Berawan
15:41 10/10/2026

5.6 Latihan

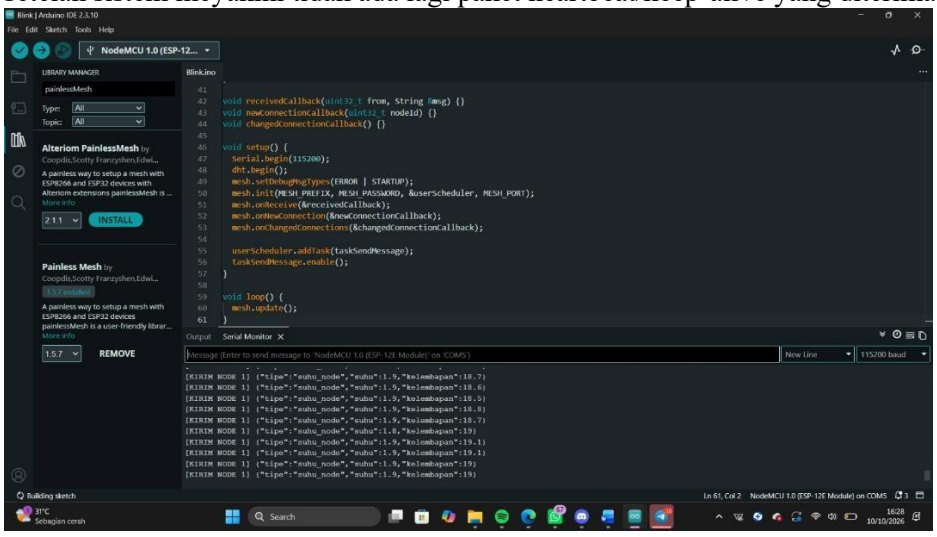
Latihan 1 (Eksperimen Deteksi ACK & Pemutusan Simpul - Praktikum 2): Saat program Praktikum 2 (ESP-NOW One-to-Many) sedang berjalan normal dan mencetak status "Berhasil Diterima" (sendStatus == 0), lakukan eksperimen pemutusan simpul:

Tampilan Output di node 1 setelah USB node 2 atau punya mita di cabut:



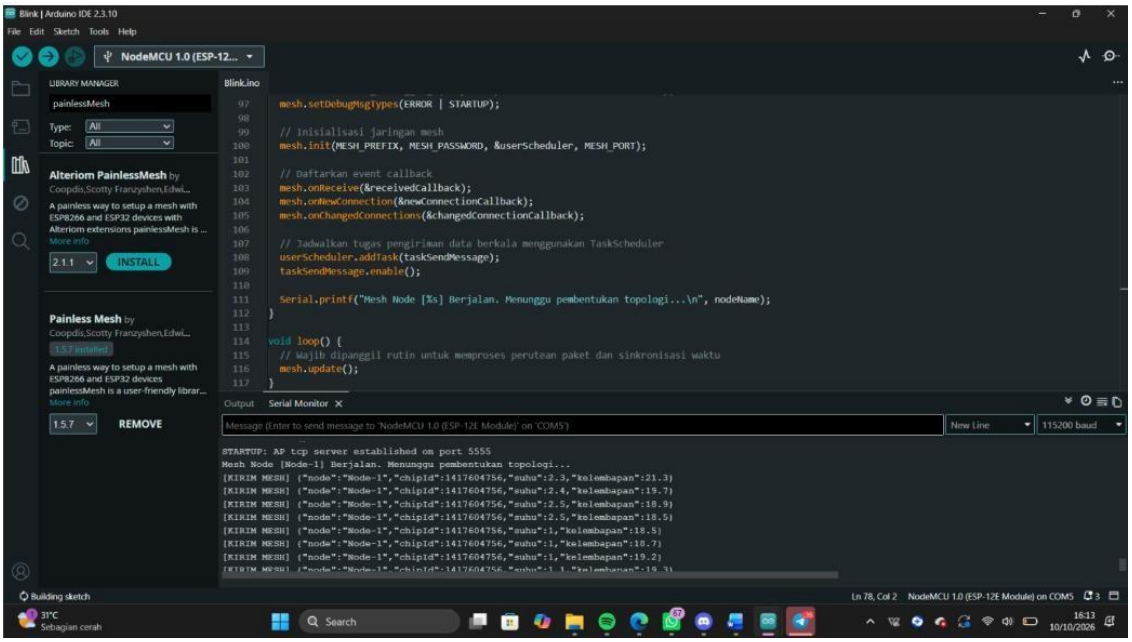
Latihan 2 (Eksperimen Dinamika Topologi & Self-Healing - Praktikum 4): Jalankan jaringan painlessMesh pada Praktikum 4 hingga seluruh board saling terhubung dan mencetak pesan --> Koneksi Baru Terdeteksi! Node ID:

Jawab:
Berikut adalah bukti dari saat saya mencabut node saya lalu saya konek an lagi tampil koneksi baru terdeteksi.
Biasanya butuh waktu sekitar 3 hingga 10 detikCallback changedConnectionCallback() akan aktif setelah sistem meyakini tidak ada lagi paket heartbeat/keep-alive yang diterima dari node tersebut.

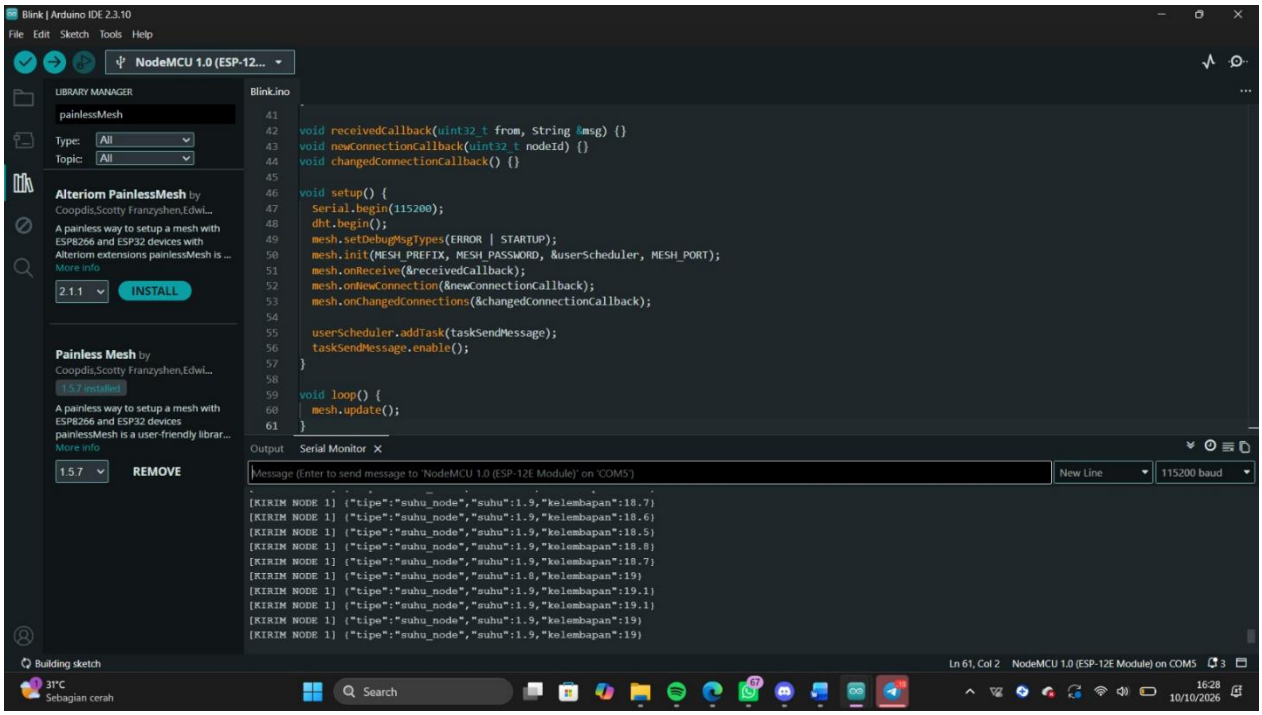


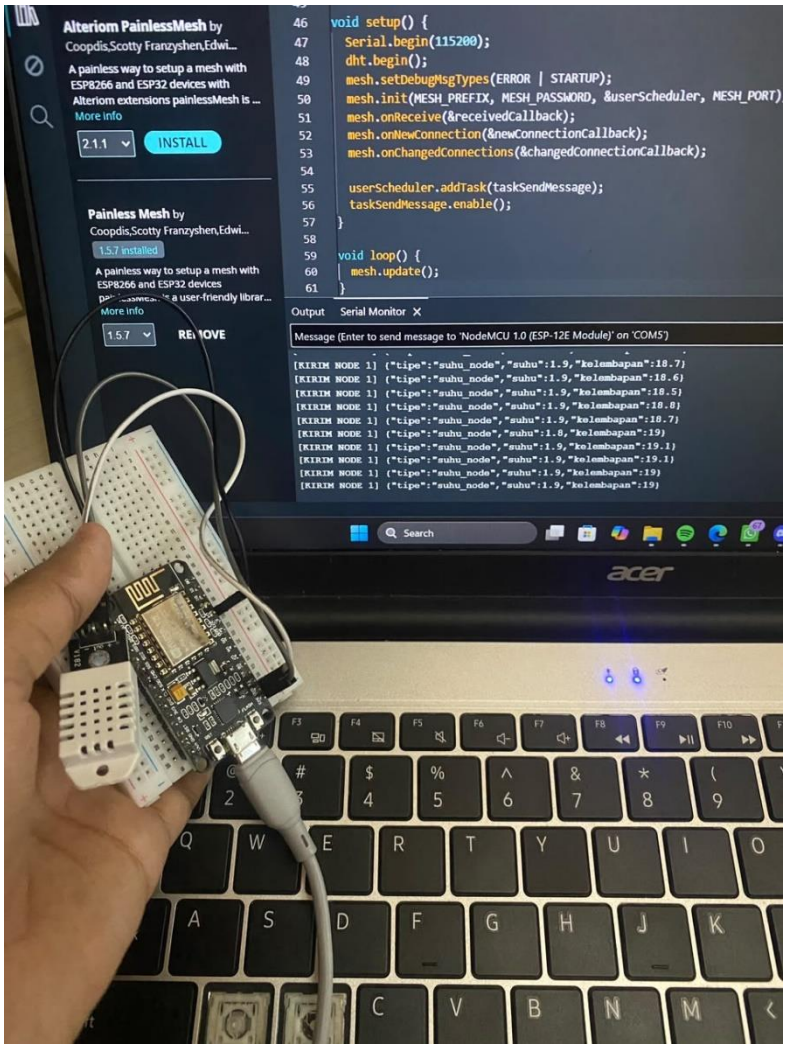
Latihan 3 (Eksperimen Modifikasi Filter Telemetri & Ambang Batas JSON - Praktikum 4): Pada Praktikum 4, setiap node saat ini mencetak seluruh data telemetri dari semua pengirim tanpa penyaringan.

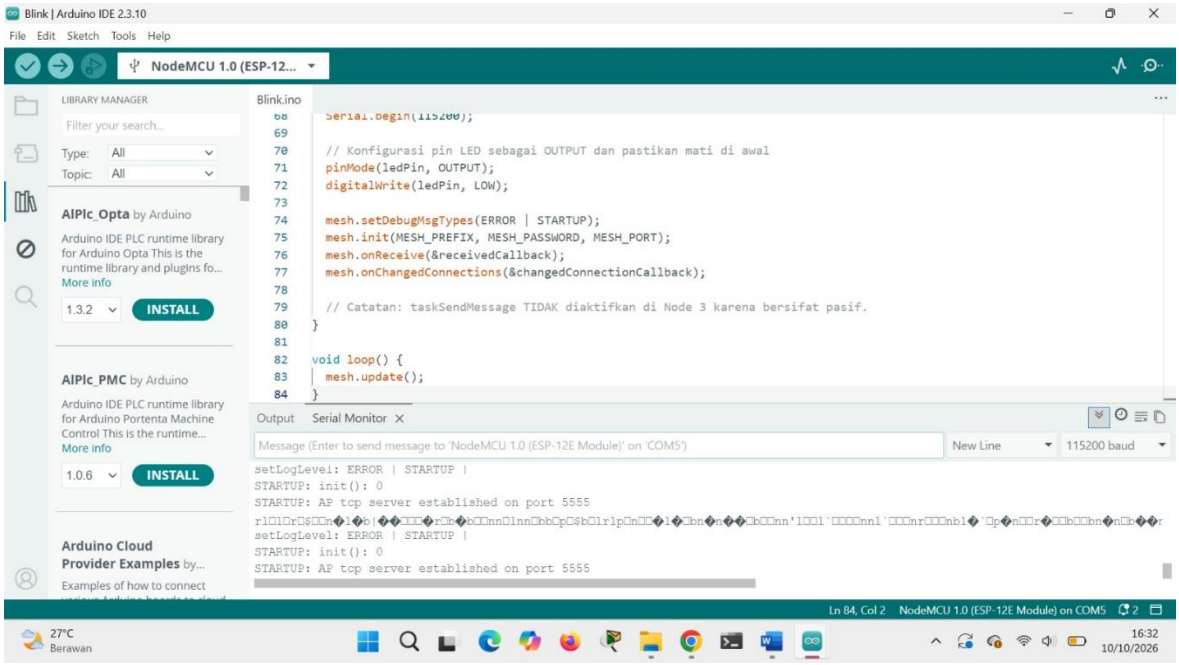
Jawab:
Berikut adalah kode dari node-1 setelah di modifikasi:

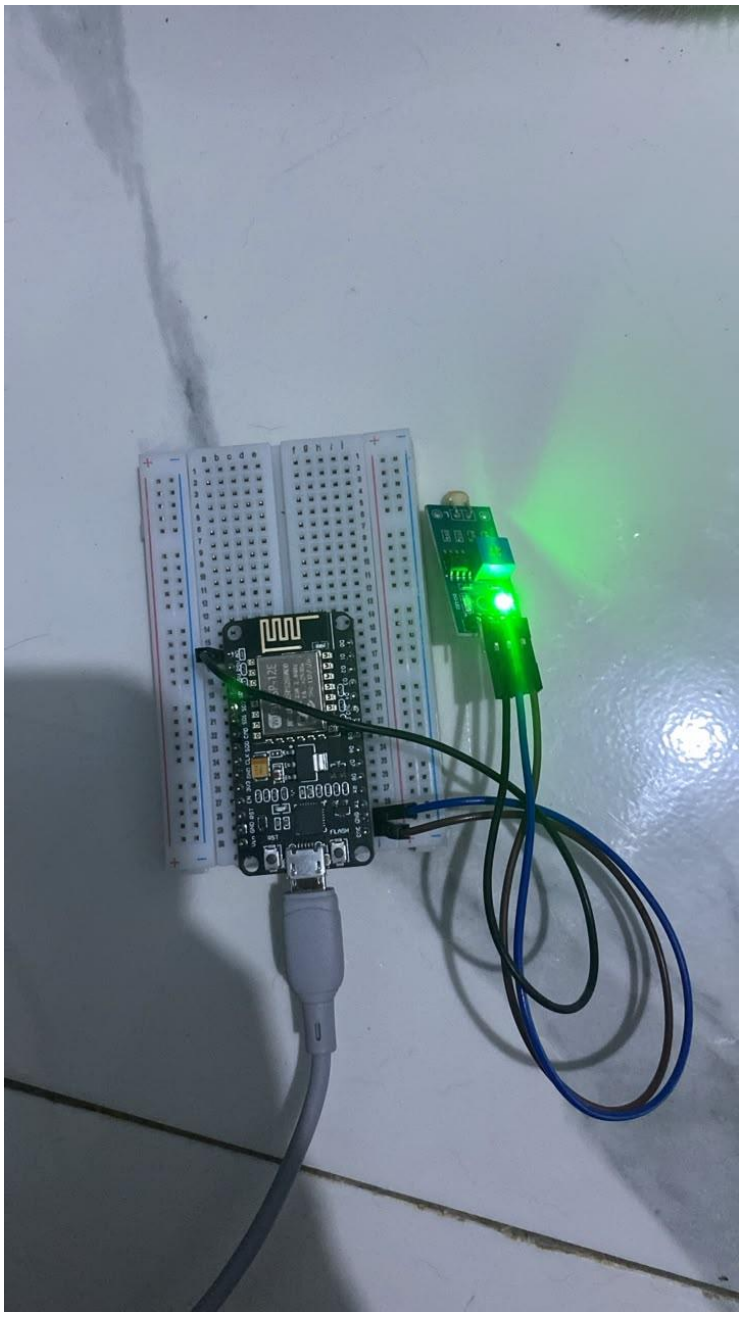


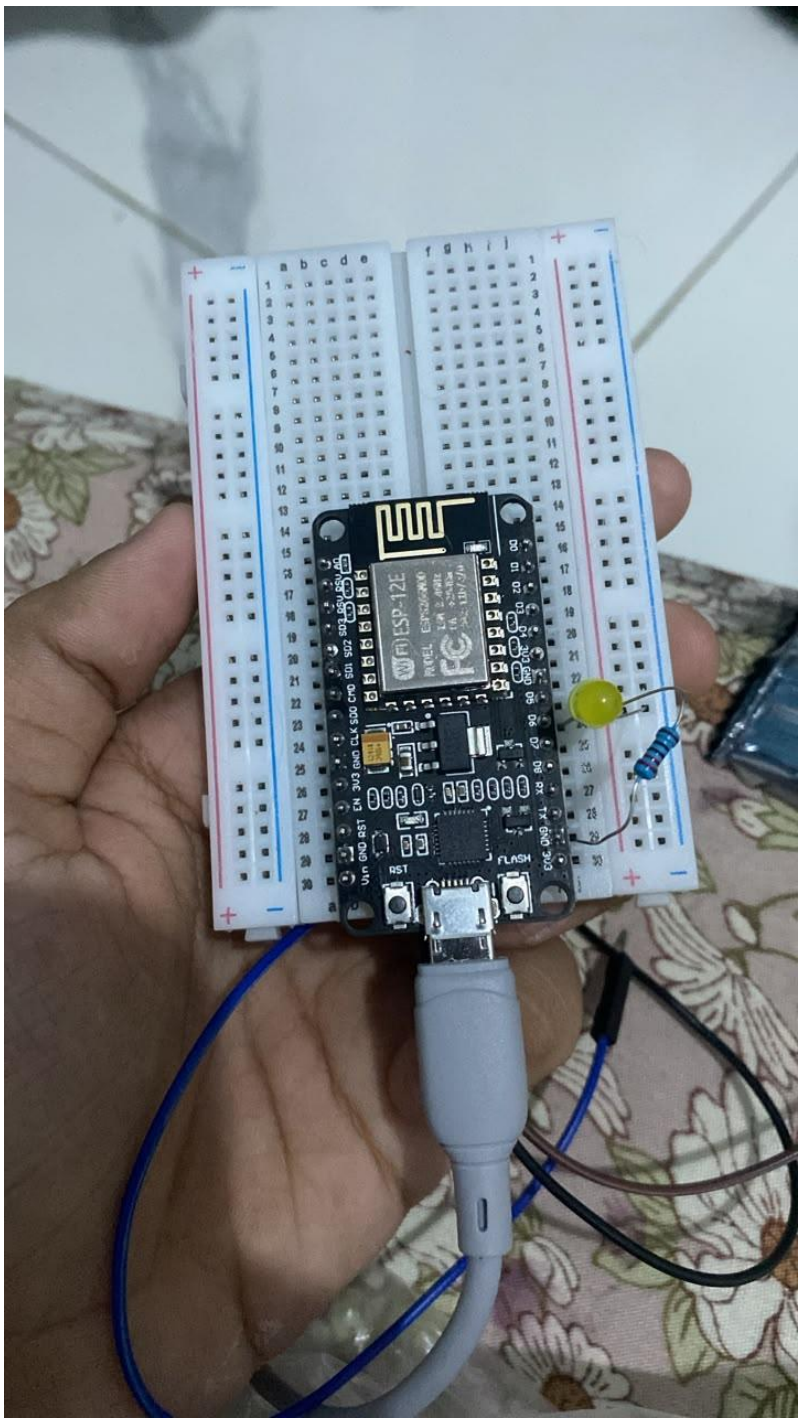
5.7 Tugas: Sistem Telemetri & Otomasi Terdistribusi Berbasis painlessMesh



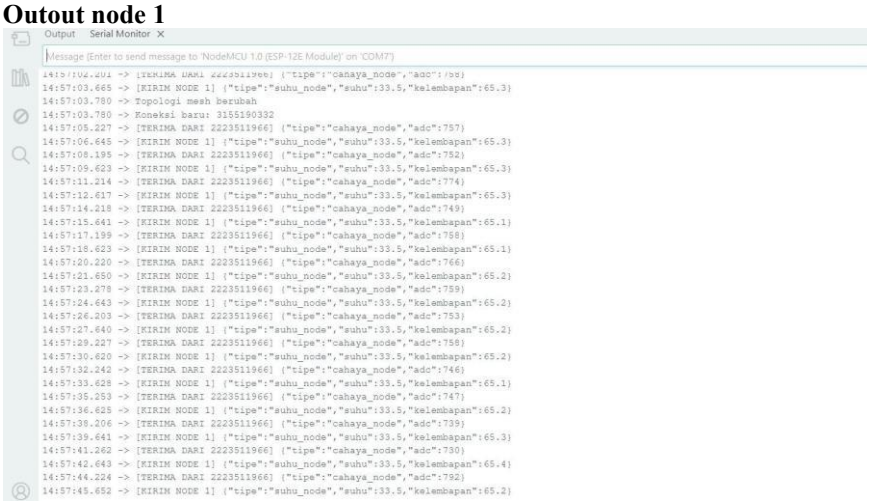



Blink | Arduino IDE 2.3.10
File Edit Sketch Tools Help
NodeMCU 1.0 (ESP-12...
LIBRARY MANAGER
Filter your search...
Type: All
Topic: All
AIPlc_Opta by Arduino
Arduino IDE PLC runtime library for Arduino Opta This is the runtime library and plugins fo...
More info
1.3.2
INSTALL
AIPlc_PMC by Arduino
Arduino IDE PLC runtime library for Arduino Portenta Machine Control This is the runtime...
More info
1.0.6
INSTALL
Arduino Cloud Provider Examples by...
Examples of how to connect
Blink.ino
Serial.begin(115200);
// Konfigurasi pin LED sebagai OUTPUT dan pastikan mati di awal
pinMode(ledPin, OUTPUT);
digitalWrite(ledPin, LOW);
mesh.setDebugMsgTypes(ERROR | STARTUP);
mesh.init(MESH_PREFIX, MESH_PASSWORD, MESH_PORT);
mesh.onReceive(&receivedCallback);
mesh.onChangedConnections(&changedConnectionCallback);
// Catatan: taskSendMessage TIDAK diaktifkan di Node 3 karena bersifat pasif.
}
void loop() {
mesh.update();
}
Output Serial Monitor
Message (Enter to send message to 'NodeMCU 1.0 (ESP-12E Module)' on 'COM5')
New Line
115200 baud
setLogLevel: ERROR | STARTUP |
STARTUP: init(): 0
STARTUP: AP tcp server established on port 5555
setLogLevel: ERROR | STARTUP |
STARTUP: init(): 0
STARTUP: AP tcp server established on port 5555
Ln 84, Col 2 NodeMCU 1.0 (ESP-12E Module) on COM5
27°C
Berawan
16:32
10/10/2026





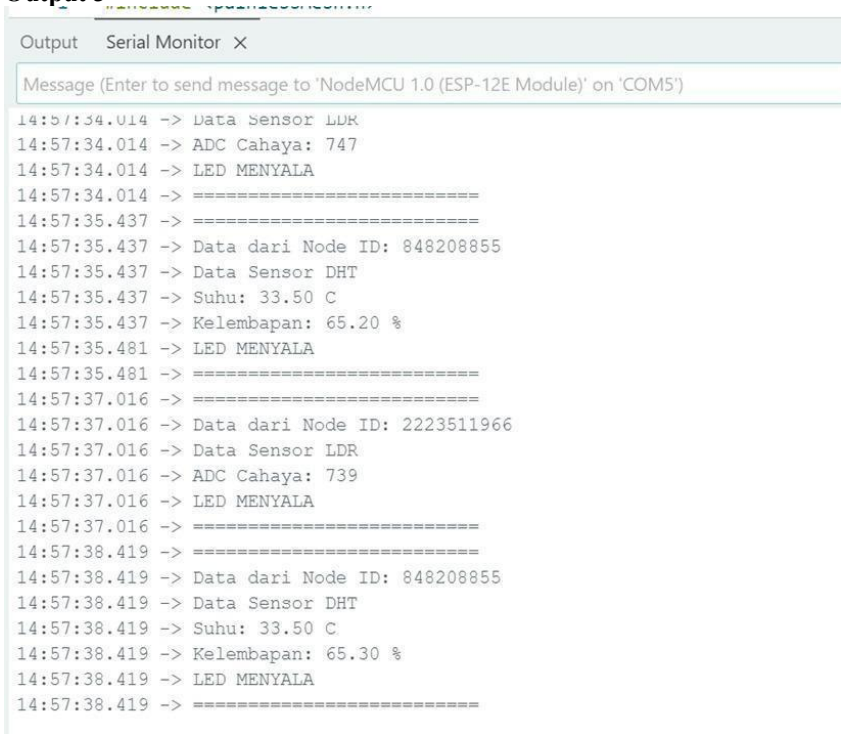
Outout node 1



Output node 2



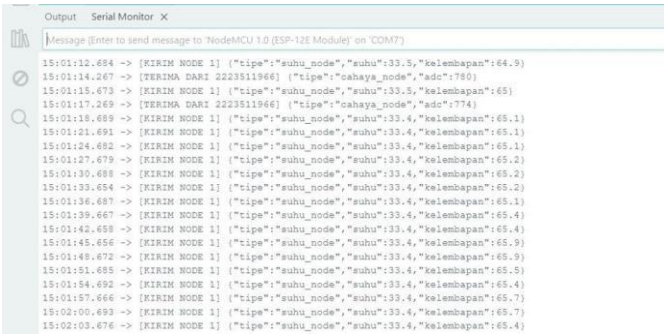
Output 3



Saat node 2 di cabut daya nya yang terjadi pada



node 1 dan 3 adalah:


Output Serial Monitor ×
Message (Enter to send message to 'NodeMCU 1.0 (ESP-12E Module)' on 'COM7')
15:01:12.684 -> [KIRIM NODE 1] {"tipe":"suhu_node","suhu":33.5,"kelembapan":64.9}
15:01:14.267 -> [TERIMA DARI 2223511966] {"tipe":"cahaya_node","adc":780}
15:01:15.673 -> [KIRIM NODE 1] {"tipe":"suhu_node","suhu":33.5,"kelembapan":65}
15:01:17.269 -> [TERIMA DARI 2223511966] {"tipe":"cahaya_node","adc":774}
15:01:18.689 -> [KIRIM NODE 1] {"tipe":"suhu_node","suhu":33.4,"kelembapan":65.1}
15:01:21.691 -> [KIRIM NODE 1] {"tipe":"suhu_node","suhu":33.4,"kelembapan":65.1}
15:01:24.682 -> [KIRIM NODE 1] {"tipe":"suhu_node","suhu":33.4,"kelembapan":65.1}
15:01:27.679 -> [KIRIM NODE 1] {"tipe":"suhu_node","suhu":33.4,"kelembapan":65.2}
15:01:30.688 -> [KIRIM NODE 1] {"tipe":"suhu_node","suhu":33.4,"kelembapan":65.2}
15:01:33.654 -> [KIRIM NODE 1] {"tipe":"suhu_node","suhu":33.4,"kelembapan":65.2}
15:01:36.687 -> [KIRIM NODE 1] {"tipe":"suhu_node","suhu":33.4,"kelembapan":65.1}
15:01:39.667 -> [KIRIM NODE 1] {"tipe":"suhu_node","suhu":33.4,"kelembapan":65.4}
15:01:42.658 -> [KIRIM NODE 1] {"tipe":"suhu_node","suhu":33.4,"kelembapan":65.4}
15:01:45.656 -> [KIRIM NODE 1] {"tipe":"suhu_node","suhu":33.4,"kelembapan":65.9}
15:01:48.672 -> [KIRIM NODE 1] {"tipe":"suhu_node","suhu":33.4,"kelembapan":65.9}
15:01:51.685 -> [KIRIM NODE 1] {"tipe":"suhu_node","suhu":33.4,"kelembapan":65.5}
15:01:54.692 -> [KIRIM NODE 1] {"tipe":"suhu_node","suhu":33.4,"kelembapan":65.4}
15:01:57.666 -> [KIRIM NODE 1] {"tipe":"suhu_node","suhu":33.4,"kelembapan":65.7}
15:02:00.693 -> [KIRIM NODE 1] {"tipe":"suhu_node","suhu":33.4,"kelembapan":65.7}
15:02:03.676 -> [KIRIM NODE 1] {"tipe":"suhu_node","suhu":33.4,"kelembapan":65.4}


Output Serial Monitor ×
Message (Enter to send message to 'NodeMCU 1.0 (ESP-12E Module)' on 'COM5')
15:02:05.565 -> Kelembapan: 65.40 %
15:02:05.565 -> LED MENYALA
15:02:05.565 -> ==========
15:02:08.398 -> ==========
15:02:08.398 -> Data dari Node ID: 848208855
15:02:08.433 -> Data Sensor DHT
15:02:08.433 -> Suhu: 33.40 C
15:02:08.433 -> Kelembapan: 65.70 %
15:02:08.433 -> LED MENYALA
15:02:08.433 -> ==========
15:02:11.434 -> ==========
15:02:11.434 -> Data dari Node ID: 848208855
15:02:11.434 -> Data Sensor DHT
15:02:11.434 -> Suhu: 33.40 C
15:02:11.434 -> Kelembapan: 65.70 %
15:02:11.434 -> LED MENYALA
15:02:11.434 -> ==========
15:02:14.485 -> ==========
15:02:14.485 -> Data dari Node ID: 848208855
15:02:14.485 -> Data Sensor DHT
15:02:14.485 -> Suhu: 33.40 C
15:02:14.485 -> Kelembapan: 65.60 %
15:02:14.485 -> LED MENYALA
15:02:14.485 -> ==========